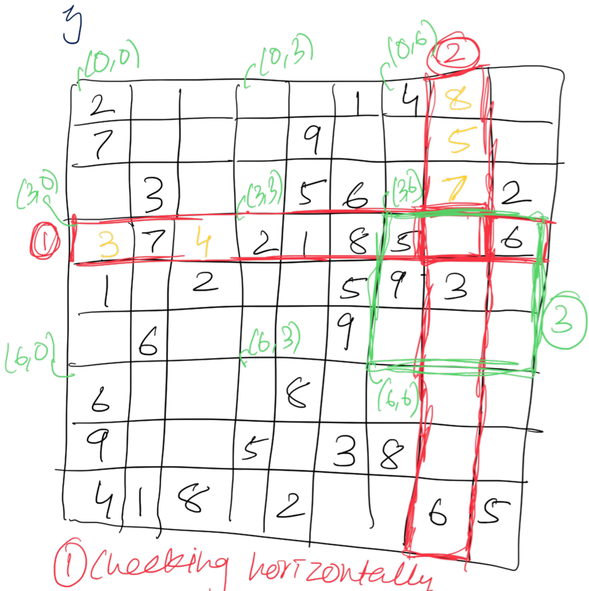
Sudoku solver

```
helper (board, row, col) {
    if (row = 9) { // when all rows are filled and we reach the last
        true
    }

    nextRow = row, nextCol = col + 1
    if (nextCol = 9) { // this is when each row wise all columns are traversed
        nextRow = row + 1;
        nextCol = 0;
    }
    if (board[row][col] != '.') { // if we get number simply go ahead
        return helper(board, nextRow, nextCol)
    }
    for (digit = 1 to 9) { // to place digit
        if ( isSafe(board, row, col, dig)) {
            board[row][col] = dig; // if right place then place
            if (helper (board, nextRow, nextCol)) {
                return true;
            }
            board[row][col] = '.'; // incorrect place so we backtrack & reset it.
        }
    }
    return false;
}
```



① Checking horizontally

```
for (j = 0 to 9) {
    if (board[row][j] == digit)
        return false
```

② Checking vertically

```
for (i = 0 to 9) {
    if (board[i][col] == digit)
        return false
```

③ Checking grid

```
for (i = sr, i <= sr + 2; i++)
    for (j = sc; j <= sc + 2; j++)
        if (board[i][j] == dig)
            return false
```

if we need to check at r=4 & c=4
go to find sr and sc use

$srow = (r/3) * 3 = (4/3) * 3 = 1 * 3 = 3$

$scol = (c/3) * 3 = (4/3) * 3 = 1 * 3 = 3$

(3, 3) take grid wise kai

TC: $O(9^{\text{empty choices}})$